

Course- B.Tech	Type- Core
Course Code- CSET106	Course Name- Discrete Mathematical Structures
Year- 2026	Semester- Even
Date- 16/03/2026	Batch- 2025-2029

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
CO1									
CO2	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO3			✓					✓	

Objectives

1. Students will be able to recognize the theory of modular arithmetic.
2. Students will be able to understand the applications related to modular arithmetic such as CRT.

Questions:

1. Let $A=\mathbb{Z}$. Define a relation R on A by
 $aRb \Leftrightarrow a \equiv b \pmod{n}$ where n is a fixed positive integer.
 Prove that R is an equivalence relation on the set of integers
2. Determine
 - (i). equivalence classes modulo 5.
 - (ii). residual classes modulo 4.
3. Let m be a positive integer and let a , b , and c be integers. Prove that
 - (i). If $\gcd(a,b) = 1$ and $a \mid bc$, then $a \mid c$.
 - (ii). If $ac \equiv bc \pmod{m}$ and $\gcd(c,m) = 1$, then $a \equiv b \pmod{m}$.
4. Determine the solution for the following system using Chinese Remainder Theorem.
 $x \equiv 3 \pmod{9}$
 $x \equiv 7 \pmod{13}$
5. Determine the solution for the following system using Chinese remainder theorem.
 $x \equiv 2 \pmod{3}$
 $x \equiv 3 \pmod{5}$
 $x \equiv 2 \pmod{7}$